FADEEL SHER KHAN

Austin, TX 78712 | (832) 679-1657 | fadeelkhan@utexas.edu | linkedin.com/in/fadeelsherkhan

EDUCATION

The University of Texas at Austin — Austin, TX

PhD, Biomedical Engineering (Focus in Applied Machine Learning) — Expected: May 2028

Master of Science in Engineering, Biomedical Engineering — Dec. 2025

Advisors: Mia Markey, James Tunnell

Rice University — Houston, TX

Bachelor of Science, Bioengineering; Minor in Data Science — May 2023

WORK EXPERIENCE

The University of Texas at Austin — Austin, TX

Graduate Research Assistant (Funded by NIH Grant: R01CA273734) — Aug. 2023 – Present

- Designing generative artificial intelligence (AI) methods to reduce the cost of tissue processing and histopathology for Mohs surgery for skin cancer treatment
- Developing state-of-the-art AI/ML models for rapid "virtual" staining using reflectance confocal microscopy, stimulated Raman scattering microscopy, fluorescence imitating brightfield microscopy, and fluorescence confocal microscopy; these methods act as alternatives to costly, slow chemical staining
- Constructing image quality assessment guidelines for the reliable and safe integration of virtual staining
- Conducting AI-based automated clinical decision-making for melanoma and basal cell carcinomas

Samsung Research America — Plano, TX

Computational Imaging Intern, Mobile Processor Innovation Team — May 2025 – Aug. 2025

- Developed a novel method for automated image quality assessment for generative AI models
- Researched and developed advanced AI algorithms using synthetic data generation, contrastive learning, reward modeling and image segmentation; global patent and research paper in development

Computational Imaging Intern, Mobile Processor Innovation Team — Jun. 2024 – Aug. 2024

- Developed a novel AI pipeline for multi-frame image-super resolution via synthetic data generation
- Researched advanced computational imaging algorithms for challenging computational photography applications such as digital zoom and low light (night) photography for Samsung Galaxy phones
- Designed light-weight AI architectures for Samsung NPUs; global patent submitted, and research paper presented at IEEE/CVF CVPR 2025 conference (see publications section below)

Rice 360 Institute for Global Health Technologies — Houston, TX

Engineering Research Intern — May. 2021 – Jul. 2021; Aug. 2022 – May 2023

- Designed, developed and prototyped a UV-based sterilization system in 3-member team for the safe reuse of N95 masks during COVID-19 pandemic in low-resource countries

- Collaborated with engineering teams in Malawi, Tanzania, and Houston to develop data science solutions to global health challenges

Computational Wellbeing Group, Rice University Houston, TX

Undergraduate Machine Learning Research Assistant Sep. 2020 – May 2021

- Implemented deep learning models using computer vision tools (*opencv, pytorch etc.*) for real-time health monitoring using web cameras (i.e., changes in emotion, stress levels, vitals like heart rate) during the COVID-19 pandemic to understand the productivity and wellbeing of participants in isolated work settings
- Aided in conducting remote clinical trials with 30 participants to generate time series wearable sensing data

PEER-REVIEWED JOURNAL PUBLICATIONS

- **Fadeel S. Khan**, Mia K. Markey, Umberto E. Villa, Alan C. Bovik, Matthew C. Fox, Brett H. Keeling, Jason S. Reichenberg, James W. Tunnell. "Context-Specific Image Quality Assessment Guideline for Virtual Histologic Staining." *SPIE Journal of Medical Imaging* 2026, 13(S11209) {*SPIE*}
- Mengkun Chen, Yen-Tung Liu, **Fadeel S. Khan**, Matthew C. Fox, Jason S. Reichenberg, Fabiana C.P.S. Lopes, Katherine R. Sebastian, Mia K. Markey, James W. Tunnell. "Single Color Digital H&E staining with In-and-Out Net." *Computerized Medical Imaging and Graphics Journal 2024, 118:102468 {CMIG}*
- **Fadeel S. Khan**, Mengkun Chen, Sanidhya Tripathi, James W. Tunnell. "Simulation-Driven Deep Learning Framework for Raman Spectral Denoising Under Fluorescence-Dominant Conditions" *Spectrochimica Acta Part A: Molecular and Biomolecular Spectroscopy (under revision)*
- Partick Giolando, Amanda Nowacki, Debarghya Chaki, **Fadeel S. Khan**, et al. "Non-Uniform Rational B-Spline-Based Deep Learning Segmentation of the Left Ventricular Cavity and Myocardium from Apical 4 Chamber Echocardiogram" *SPIE Journal of Medical Imaging (under review)*

PEER-REVIEWED CONFERENCE PROCEEDINGS

- **Fadeel S. Khan**, Joshua P. Ebenezer, Seok-Jun Lee, Hamid R. Sheikh. "MFSR-GAN: Multi-Frame Super-Resolution with Handheld Motion Modeling." *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR) Workshops 2025 {IEEE/CVF CVPR}*
- **Fadeel S. Khan**, Xu Feng, Matthew C. Fox, Jason S. Reichenberg, Fabiana C. P. Lopes, Mia K. Markey, James W. Tunnell. "Biophysically Interpretable Basal Cell Carcinoma Classification Using Raman Spectroscopy Transformer Model." *SPIE Photonics West Conference 2025 {SPIE}*
- **Fadeel S. Khan*,** Maryam Khalid*, John Broussard, Arko Barman. "Graph-based biomarker discovery and interpretation for Alzheimer's disease." *IEEE International Conference on Healthcare Informatics (ICHI) 2026* [*equal first author contribution] {*IEEE*}
- Brian K. Lee, Logan B. Parker, Alex T. Whitman, **Fadeel S. Khan**, Matthew C. Fox, Jason S. Reichenberg, Brett H. Keeling, Katherine R. Sebastian, Mia K. Markey, James W. Tunnell. "Line-scan Fluorescence and Raman Confocal Microscope for Skin Cancer Surgical Guidance." *SPIE Photonics West Conference 2026 {SPIE}*

- **Fadeel S. Khan**, Long N. Le, Abhinau K. Venkataramanan, Seok-Jun Lee, Hamid R. Sheikh. "Pixel Perfect: Relational Image Quality Assessment with Spatially-Aware Distortions." *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR) Workshops 2026 (under review)*

PEER-REVIEWED CONFERENCE PRESENTATIONS

- **Fadeel S. Khan**, Joshua P. Ebenezer, Seok-Jun Lee, Hamid R. Sheikh. "MFSR-GAN: Multi-Frame Super-Resolution with Handheld Motion Modeling." *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR) Workshops 2025 Poster Presentation*
- **Fadeel S. Khan**, Xu Feng, Matthew C. Fox, Jason S. Reichenberg, Fabiana C. P. Lopes, Mia K. Markey, James W. Tunnell. "Biophysically Interpretable Basal Cell Carcinoma Classification Using Raman Spectroscopy Transformer Model." *SPIE Photonics West Conference 2025 Oral Presentation*
- **Fadeel S. Khan,** Sam Freer, Mia Markey, Sapun Parekh, Matthew Fox, James Tunnell. "The Case for Context-Specific Image Quality Assessment for Virtual Histology." *Medical Image Perception Society (MIPS) 2026*

PATENTS UNDER DEVELOPMENT

- **Fadeel S. Khan**, Joshua P. Ebenezer, & Hamid R. Sheikh. *End-to-End Multi-frame Image Super-Resolution via Synthetic Data Engine and Artificial Intelligence Methods.* (patent pending)
- **Fadeel S. Khan**, Long N. Le, & Hamid R. Sheikh. *Automated Image Quality Assessment Using Contrastive Learning for Mobile Personalization Applications.* (patent application in development)

PEER REVIEW

- Peer reviewer for IEEE/CVF CVPR, IEEE ICIP, IEEE OJSP

RESEARCH PROJECTS

Biomedical Informatics & Biophotonics Laboratories, The University of Texas at Austin Austin, TX

AI-Based Virtual Melanoma Staining Using Microscopy Sept. 2024 – Present

- Primary PhD thesis project focusing on the generation of multiplexed virtual histopathology stains from unstained microscopy images for low-cost and rapid diagnostic assessment of melanoma during surgery
- Successful implementation of this project is estimated to reduce medical costs associated with skin cancer surgery by 80% by replacing "chemical" staining of tissue samples with digital AI-based alternatives

AI-Based Surgical Guidance for Non-Melanoma Skin Cancer Dec. 2023 – Present

- Creating co-registered multimodal datasets and AI-based analysis pipelines for multimodal optics (fluorescence/reflectance confocal, Raman scattering microscopies) for non-melanoma skin cancer visualization, detection, and surgical guidance

Medical Image Quality Assessment in Digital Pathology Jun. 2023 – Present

- Collaborating with computational researchers and clinicians to create guidelines for appropriate usage of image quality assessment metrics in digital pathology to guide translation of research into clinical practice

Myocardium Image Segmentation Aug. 2023 – Dec. 2024

- Designed regression models to perform image segmentation of myocardium using noisy echocardiograms; conducted data labelling & augmentation, model evaluation, and clinical model validation

Texas Alzheimer's Research and Care Consortium (TARCC) Houston, TX

Machine Learning Diagnosis of Alzheimer's Disease Biomarkers Aug. 2022 – May 2023

- Created machine learning pipeline and graph-based explanations for predicting early Alzheimer's Disease
- Conducted feature engineering, feature selection and model interpretation for ensemble models

Rice University Houston, TX

Senior Design Project: Fall and Activity Sensing Using Machine Learning Aug. 2022 – May 2023

- Developed waist-wearable sensing device using 9 degrees-of-freedom accelerometer, LiDAR scanner and ultrawideband sensors; conducted prototyping, experimentation and clinical validation
- Created machine learning ensemble models for on-device real-time fall and activity classification
- Collaborated with clinicians to create highly usable interface for real-time fall and activity monitoring

TEACHING EXPERIENCE

GLHT 201: Intro. to Global Health Technologies, Rice University Houston, TX

Undergraduate Teaching Assistant Aug. 2021 – Dec. 2021

- Mentored undergraduate teams for medical device design hackathon to conduct engineering design and prototyping; led class discussions on health equity, innovation in global health, and new-born health
- Delivered mock debates for class and facilitated regular class debates on various topics in global health

LEADERSHIP EXPERIENCE

IEEE

Student Member Dec. 2023 – Present

SPIE

Student Member Dec. 2023 – Present

BMES

Student Member Dec. 2023 – Present

Rice University Biomedical Engineering Society (BMES)

President Aug. 2021 – May 2023

- Oversaw all club affairs, and liaise with external groups, Rice faculty, administration and student leadership
- Negotiated with administrative leadership and external organizations to secure club funding and sponsorship deals

SERVICE EXPERIENCE

Partners in Health (PIH) Engage at Rice

President (previously Fundraising Lead) Oct. 2019 – Aug. 2023

- Led global health activities on campus including organizing fundraising, raising awareness and public

advocacy

- Raised $4,800 from 135+ donors as lead of 15-person team by organizing on-campus and virtual fundraising events to support HIV prevention in Haiti, COVID relief in multiple countries, and maternal and infant healthcare in Sierra Leone
- Campaigned for US health policy change towards fighting TB, and COVID; met with offices of House Rep. Fletcher and Senator John Cornyn; persuaded Rep. Fletcher to sign legislation (Global TB "Dear Colleague" letter) to support global health funding by Congress

SKILLS

Computational: Python (*pytorch, numpy, opencv, pandas*), SQL, MATLAB, Linux/Unix, AWS, R

Engineering: Optics and Microscopy, Prototyping, LabVIEW, Arduino, CAD (*AutoCAD, SolidWorks*), Circuit Design, 3D Printing, Laser Cutting

Wet Lab: Tissue Culture, Molecular Cloning Techniques, Bacterial Cell Culture, Bioreactors